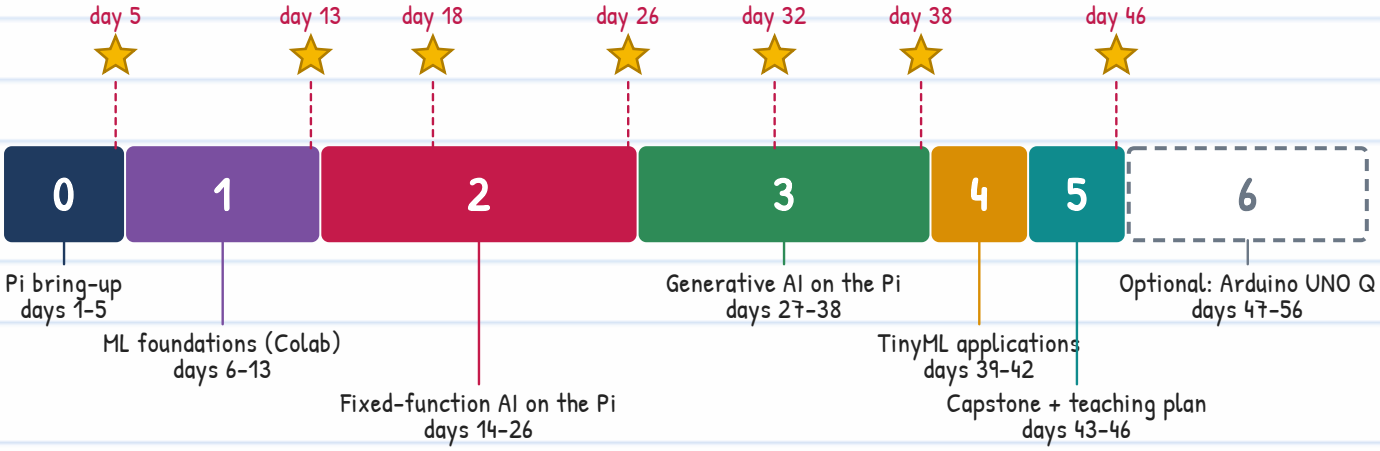
46 DAYS, SIX PHASES

From absolute beginner to Edge AI engineer: about 2 hours a day, roughly nine weeks at five days a week, on a Raspberry Pi 4 or 5. Optional extension: Arduino UNO Q.

1. The course in one picture



Stars = what I can demonstrate: day 5 Pi ready · day 13 first CNN · day 18 own classifier on the Pi · day 26 object counting · day 32 RAG · day 38 a language model drives GPIO · day 46 capstone.

my notes

2. The phases

Phase	Days	What I learn	What I need
0 Pi bring-up	1–5	Flash, boot, remote access, Linux basics, camera, Python and Jupyter on the Pi	Raspberry Pi 4 or 5, microSD, power supply, laptop
1 ML foundations (Colab)	6–13	Neurons, deep neural networks, CNNs, datasets, metrics, overfitting, Edge Impulse first steps	Laptop and Google Colab; the Pi is not needed
2 Fixed-function AI on the Pi	14–26	Image classification, object detection (SSD, FOMO), YOLO, counting objects	Pi camera or USB webcam; Edge Impulse account
3 Generative AI on the Pi	27–38	RNN text, distillation, small language models, RAG, tools, agents, vision-language, audio, GPIO, IoT control	USB microphone and speaker; breadboard, LEDs, button, sensors
4 TinyML applications	39–42	Anomaly detection, keyword spotting, image applications on microcontrollers, responsible AI	Concepts only: no microcontroller board needed
5 Capstone + teaching plan	43–46	Your own project, the write-up, and the plan to teach it to students	Whatever the project needs
6 Optional: Arduino UNO Q	47–56	The same ideas on a second hardware platform, after the Pi and ML are mastered	Arduino UNO Q board (optional)

my notes

HOW THE COURSE WORKS

3. The rules of the game

- **One day = about 2 hours.** Every day has a full tutorial (complete in itself: no book or website needed), a narrated animation where it helps, hands-on steps, a quiz, and notebook notes with space for my own comments.
- **Pi 4 and Pi 5:** every day works on both boards. Where they differ (speed, memory, power, camera cable, model sizes) the lesson says so.
- **Learn it twice:** with two Pis, learn on one, then repeat on the other from my notes only. Different host names (`raspi-lab` , `raspi-home`), same user, code and notes in a shared online repository.
- **Phase 1 needs no Pi**, only Google Colab: do it anywhere.
- Day sizes are estimates; a long day simply continues the next day.

my notes

4. Notes on hardware and software

Speed and memory: vision runs well on both boards. In the language-model phase a Pi 5 (8 GB best) runs larger models; a Pi 4 uses smaller models and longer waits.
Python: a newer Raspberry Pi OS may ship a newer Python than some ML packages support; days 4–5 include the check and the remedy.
Online services (Colab, Edge Impulse, model downloads) need Internet; use a phone hotspot if the network restricts them.
UNO Q is an optional extension.

my notes

5. Get ready (by day)

- [ ] **day 1** Raspberry Pi 4 or 5, microSD (32 GB or more), the matching USB-C power supply, card reader, Wi-Fi details
- [ ] **day 4** Pi camera module (check the camera cable: it differs between Pi 4 and Pi 5) or a USB webcam
- [ ] **day 6** Google account
- [ ] **day 13** Edge Impulse account (check the current free tier)
- [ ] **day 27** Internet on the Pi for model downloads; enough free space on the card
- [ ] **day 35** USB microphone and speaker
- [ ] **day 36** Breadboard, jumper wires, LEDs with resistors, push button, DHT temperature sensor, BMP280 pressure sensor (confirm the exact parts in the lessons before buying)
- [ ] **day 47+** Arduino UNO Q (optional)

my notes

DAY BY DAY (1)

Phase 0: Pi bring-up (days 1-5)

Day	Topic	What you do	You will have
1	Meet the Raspberry Pi; flash the microSD; first boot and SSH	Learn Edge AI and the Pi 4 / Pi 5; write the OS to a card; boot; log in from your laptop	A Pi you can log into remotely
2	Remote access in depth	Copy files both ways (scp, SFTP); open the desktop remotely; add a second Wi-Fi network; work with more than one Pi	Files and desktop reachable from your laptop
3	The Linux command line	Navigate, create, edit and delete files; permissions; processes; monitoring with htop	Confidence in the terminal
4	Camera, swap memory and updates	Connect a Pi camera or USB webcam; capture a photo; adjust swap; update the system	First photo captured on the Pi
5	Python environment and Jupyter on the Pi	Create a virtual environment; install packages; run Jupyter remotely	A notebook running on the Pi

my notes

Phase 1: ML foundations (Colab) (days 6-13)

Day	Topic	What you do	You will have
6	Google Colab, Python refresher, what TinyML is	Run notebooks in the cloud; revise Python and NumPy	A working Colab setup
7	The machine-learning paradigm	Fit a first model from data and compare it with rule-based code	Your first trained model
8	Neural-network building blocks	Neurons, layers, activations, loss, gradient descent; build a small network	A small network trained from scratch
9	Regression with a deep neural network	Predict a number from data; plot predictions	A regression model
10	Classification with a deep neural network	Predict classes; read a confusion matrix	A classifier with metrics
11	Datasets, splits and metrics	Train / validation / test splits; accuracy, precision, recall, F1	Correct model evaluation
12	Convolutions; image classification with a CNN	Filters and pooling; build and train a CNN	A CNN trained in Colab
13	Edge Impulse first project; preventing overfitting	Train in the browser; dropout and data augmentation	A model ready to export

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DAY BY DAY (2)

Phase 2: Fixed-function AI on the Pi (days 14-26)

Day	Topic	What you do	You will have
14	TensorFlow Lite and the first inference on the Pi	Install TensorFlow Lite; run a pre-trained image classifier	An image classified on the Pi
15	Live camera classification; a model trained from scratch	Classify live camera frames; run a small CIFAR-10 model	Live classification
16	Custom classifier: goal and data collection	Define the problem; collect and organise your own images	Your own dataset
17	Training in Edge Impulse	Impulse design, pre-processing, training, accuracy versus speed	A trained, quantised model
18	Test, deploy and run live on the Pi	Test the model; deploy it; run live inference	Your own classifier on the Pi
19	Object detection fundamentals	Bounding boxes, labels, post-processing; a pre-trained detector	Detection on still images
20	Detection on a live stream	Run a detector on the camera stream	Live object detection
21	Custom detection: data and labelling	Collect and label images for detection	A labelled detection dataset
22	SSD MobileNet in Edge Impulse; deployment	Train, deploy and post-process an SSD model	A detector on the Pi
23	FOMO: lightweight detection	Train and deploy a FOMO model	FOMO running on the Pi
24	YOLO with Ultralytics on the Pi	Run YOLO; export to an efficient format	YOLO inference on the Pi
25	YOLO: custom training and deployment	Train YOLO on your data; deploy it	A custom YOLO model
26	Counting objects; Phase 2 mini-project	Build a counting pipeline; complete a small project	A counting system

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DAY BY DAY (3)

Phase 3: Generative AI on the Pi (days 27-38)

Day	Topic	What you do	You will have
27	Text generation with an RNN	Tokens, embeddings, training a small text generator	A tiny text generator
28	Knowledge distillation	Teach a small student model from a larger teacher	A distilled model
29	Small language models: first model with Ollama	Install Ollama; choose a model that fits your board; chat	A language model answering on the Pi
30	Ollama with Python	Use the Python library; build a chat script	A Python chat program
31	Function (tool) calling	Let the model call Python functions	A tool-using model
32	Retrieval-augmented generation (RAG)	Build a vector store; retrieve context for answers	Question answering over your documents
33	Agents with small models	Design a tool set; build a simple agent loop	A working agent
34	Vision-language models	Run a vision-language model (Florence-2) for captioning and detection	Image captioning on the Pi
35	Audio and vision AI pipeline	Microphone to speech-to-text to language model to speech	Speech in, answer out
36	GPIO and physical computing: LEDs and buttons	Wire a circuit; control it from Python	An LED and button under code control
37	Sensors and notebooks	Read temperature, humidity and pressure sensors; log data	Sensor data logged
38	Small language models for IoT control	Let a language model switch real hardware	Language-model-driven hardware

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DAY BY DAY (4)

Phase 4: TinyML applications (days 39-42)

Day	Topic	What you do	You will have
39	Anomaly detection and K-means	Cluster data; detect anomalies	An anomaly detector (concept + notebook)
40	Keyword spotting (concepts)	Audio features and the keyword-spotting pipeline	The pipeline understood
41	Image applications on microcontrollers; data collection	Person detection concept; ways to collect data	A data-collection plan
42	Responsible AI; going further with Edge AI	Fairness, privacy, safety, lifecycle	A responsible-deployment checklist

my notes

Phase 5: Capstone + teaching plan (days 43-46)

Day	Topic	What you do	You will have
43	Choose and plan the project	Define the goal, data, model and hardware	A one-page project plan
44	Build	Implement the system	A working prototype
45	Build, test and measure	Measure accuracy, speed and resource use honestly	Measured results
46	Present, write up, plan how to teach it	Report, demo, lesson plan	Report + teaching plan

my notes

Phase 6: Optional: Arduino UNO Q (days 47-56)

Day	Topic	What you do	You will have
47	UNO Q: foundations and setup	Understand the board; set it up	Board working
50	Language models on the UNO Q; benchmarking; Python	Run and benchmark models; script them	Model running
53	Multimodal and agentic AI on the UNO Q	Combine vision, language and tools	An agent running
55	Physical AI: the two processors working together	Connect a language model to hardware through the bridge	A language model driving hardware

my notes

CHECKPOINTS & REFERENCES

What I can demonstrate

- [ ] day 5: Pi ready
- [ ] day 13: first CNN
- [ ] day 18: own classifier on the Pi
- [ ] day 26: object counting
- [ ] day 32: RAG
- [ ] day 38: language model drives GPIO
- [ ] day 46: capstone

my notes

Acknowledgments and references

This course is a compilation: it combines material from the sources below with new explanations, examples, diagrams, exercises and comments written for this course. The original authors are gratefully acknowledged.

[1] Rovai, M. (2025). Edge AI Engineering: Hands-on with the Raspberry Pi. Open textbook for the course IESTI05, Federal University of Itajubá (UNIFEI), Brazil. Version of 23 October 2025. Online edition: https://mjrovai.github.io/EdgeML_Made_Ease_ebook/

[2] Rovai, M. (2026). IESTI05 - Edge AI: Edge Machine Learning Systems Engineering. Course slides, notebooks and video lectures (second semester 2026), UNIFEI, Brazil. Course videos listed by the author: Class 1 (course introduction) https://youtu.be/QAb3MFHwe78 ; Class 2 (Introduction to Linux) https://youtu.be/QkIHILtl0h4 ; Class 3 (Raspberry Pi setup walkthrough) https://youtu.be/YtjsM36LzVE (lectures in Portuguese).

[3] Rovai, M. (2026). IESTI01 - TinyML: Machine Learning for Embedding Devices. Course materials (first semester 2026), UNIFEI, Brazil. https://github.com/Mjrovai/UNIFEI-IESTI01-TinyML

[4] Rovai, M. TinyML4D: lectures, books, papers and tutorials for embedded machine-learning education. https://github.com/Mjrovai/TinyML4D

[5] Raspberry Pi Imager: download page and software. https://www.raspberrypi.com/software/

[6] Raspberry Pi Ltd. Raspberry Pi documentation (hardware specifications of the Raspberry Pi 4 and Raspberry Pi 5, operating system, remote access). https://www.raspberrypi.com/documentation/

my notes